Windows Ready Print & Windows Protected Print Mode (WPP) — Intune Deployment, Compatibility Assessment & Legacy Print-Server Migration

Implementation Solution Guide

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Executive Summary

The situation

Windows printing is changing. Microsoft's modern, **driverless** print platform — **Windows Ready Print** — replaces vendor V3/V4 printer drivers with open standards (**IPP, Mopria, eSCL, Universal Print**), and its hardened form, **Windows Protected Print mode (WPP)** on Windows 11 24H2, forces the PC to print **only** through that platform and **blocks third-party drivers**. Microsoft has also announced **end-of-servicing for legacy third-party drivers** — this is the strategic direction, not an optional feature.

Why it matters

Third-party print drivers and Point-and-Print are the attack surface behind the PrintNightmare class of Windows Print Spooler vulnerabilities. WPP closes that surface and removes a recurring source of Windows-Update breakage. But **enabling WPP removes incompatible printers and their drivers** — so done blindly, fleet-wide, it breaks printing and scanning in production. Done right, it's a clean security and reliability win that gets you ahead of Microsoft's driver retirement.

The approach

This solution deploys and hardens Windows Ready Print through Intune, safely:

1. **Assess** — inventory every driver and printer; classify each as **Ready-Print eligible**, **needs Universal Print, or** blocker** (non-Mopria, legacy software printer, eSCL-less scanner).
1. **Design** the target path — direct **IPP/Mopria** and/or **Universal Print** — and set the WPP scope.
2. **Configure** the Intune **Custom OMA-URI** profile (`ConfigureWindowsProtectedPrint = <enabled/>`), scoped to 24H2+ devices.
1. **Pilot and validate** — confirm WPP is on, printing works via the IPP class driver, and only the planned printers were removed.
1. **Migrate** off legacy print servers/Point-and-Print to the modern path.
2. **Roll out in rings**, keep non-compliant devices as tracked exceptions, and **prevent regression**.

What you get

• A click-by-click **deployment runbook** (assessment → configure → pilot → migrate → roll out).

• **Five PowerShell tools** — driver inventory, Mopria/IPP readiness, scan (eSCL) readiness, WPP-state check, and post-deployment validation.

• **Testing, Troubleshooting, and Best-Practice** guidance, with GCC High notes inline.

The outcome

A hardened, driverless print estate on Windows 11: reduced attack surface, fewer update-driven print outages, and alignment with Microsoft's direction — deployed on evidence, with the printers WPP would remove found **before** it removes them. One fixed price; re-run the assessment and validation as the fleet and printer estate evolve.

Solution Overview

What this solution is

A complete, Intune-managed **deployment strategy for Windows Ready Print and Windows Protected Print mode (WPP)** on Windows 11 — the move from legacy, vendor-driver printing to Microsoft's modern, driverless print architecture, and then hardening it with Windows Protected Print mode. It takes you from a **printer/driver compatibility assessment**, through the **Intune configuration**, a **pilot**, and a **migration off legacy print servers**, to **broad rollout, validation, and prevention of regression** — with the risk of enabling WPP fleet-wide managed deliberately rather than discovered in production.

The two things, and why they're different

Term	What it is
Windows Ready Print	Microsoft's modern **driverless** print platform. Windows talks to printers over open standards — **IPP** (Internet Printing Protocol), **Mopria**-certified printing, **eSCL** for scanning, and **Universal Print** for cloud printing — using the built-in **Microsoft IPP Class Driver** instead of a vendor `.INF`/V3/V4 driver package.
Windows Protected Print mode (WPP)	A **Windows 11 security mode** that forces the PC to use Windows Ready Print **exclusively** and **blocks third-party print drivers** from loading. It's the *hardened* form of the modern platform.

Windows Ready Print is the platform; **WPP is the switch that enforces it and shuts the door on third-party drivers** — the attack surface behind PrintNightmare-class Print Spooler and Point-and-Print vulnerabilities.

Why Microsoft is moving here (and why it matters to you)

Microsoft has announced an **end-of-servicing strategy for legacy V3/V4 third-party Windows printer drivers**. Windows Ready Print isn't just another feature — it's Microsoft's **strategic direction** for Windows printing. The payoff is threefold: **security** (no arbitrary third-party drivers), **reliability** (Windows Updates are far less likely to break a vendor driver), and **architecture independence** (the same model works on x64 and ARM). For an Intune-managed fleet, adopting it now is getting ahead of a transition that is coming regardless.

The architecture, three ways

```
Legacy
```

```
  Print Server -> Point and Print -> Vendor V3/V4 Driver -> Windows Client

Modern
  IPP / Mopria Printer  or  Universal Print -> Windows Ready Print -> Windows Client

Hardened Modern
  IPP / Mopria Printer  or  Universal Print -> Windows Ready Print + Windows Protected Print
      -> Intune-managed Windows 11 Client
```

What happens when you enable WPP — the part to respect

Enabling Windows Protected Print mode is **not** a silent, cosmetic change. On an enabled device:

- Printers using **unsupported third-party drivers are removed**, and their drivers are **removed from the usable driver store**.
- A **Mopria-certified** printer can normally be **reinstalled** via Windows Ready Print (IPP class driver) instead.
- **Point and Print can no longer install third-party drivers.**
- **Some legacy "software" printers** (vendor PDF/"print to" utilities) may be removed.
- **Some multifunction scanning may stop working** unless the scanner supports the required **Mopria/eSCL** standards.

WPP is a **device-scoped** setting (Computer Configuration) — it affects **every user** on the PC, which matters for shared/kiosk devices. **This is why WPP is never enabled fleet-wide without a compatibility assessment first** — that assessment is the heart of this solution.

How it's delivered through Intune

Microsoft exposes WPP through the **Policy CSP**:

`./Device/Vendor/MSFT/Policy/Config/Printers/ConfigureWindowsProtectedPrint`

It applies to **Windows 11 24H2 and later** (Pro, Enterprise, Education, IoT Enterprise) and maps to the GPO **Computer Configuration** → **Printers** → **Configure Windows protected print**, backing the registry value `WindowsProtectedPrintGroupPolicyState` under `HKLM\Software\Policies\Microsoft\Windows NT\Printers\WPP`. As of this writing the setting is **not yet in the Intune Settings Catalog**, so it is deployed with a **Custom OMA-URI** configuration profile — this guide gives the exact profile.

The deployment model

Assess (drivers, printers, Mopria/IPP + scan readiness) → **Design the target (Ready Print / Universal Print; identify blockers)** → **Configure the Intune Custom OMA-URI profile** → **Pilot on a ring and validate** → **Migrate off legacy print servers** → **Roll out in rings** → **Validate and prevent regression.** Every step is evidence-led, and the included scripts turn compatibility and validation into yes/no answers.

Benefits

- **Security** — closes the third-party driver / Point-and-Print attack surface on hardened devices.
- **Reliability** — driverless printing that Windows Updates don't break.
- **A safe path** — the compatibility assessment finds what WPP would remove *before* it removes it.
- **Strategic alignment** — ahead of Microsoft's V3/V4 driver end-of-servicing.

Limitations & scope

- Scope is **deploying and hardening Windows Ready Print / WPP through Intune** on **Windows 11 24H2+**.

Printer-side enablement (turning on IPP/IPPS, Mopria certification) is vendor-specific and referenced, not rebuilt.

- **Sovereign clouds:** the CSP and method are identical in GCC / GCC High / DoD (portal

`intune.microsoft.us`); **Universal Print availability differs by sovereign cloud** — confirm for yours. Called out inline; a single deliverable covers both.

Requirements

Confirm each prerequisite before you begin. Each lists **why** it is required, **how to verify**, and **how to implement** if missing.

1. Windows 11 24H2 or later on the target devices

- **Description:** Windows Protected Print mode and the `ConfigureWindowsProtectedPrint` CSP require

Windows 11, version 24H2 or later — **Pro, Enterprise, Education, or IoT Enterprise**.

- **Why required:** The setting does not exist on earlier builds; targeting them is a no-op.
- **How to verify:** `winver`, or `Get-ComputerInfo -Property OsName,OsBuildNumber` (24H2 = build 26100+).

The included `WPPRINT-Get-WPPState.ps1` reports the build and eligibility.

- **How to implement if missing:** Update devices to 24H2 first; scope WPP only to eligible devices (use

an Intune **filter** on OS version).

2. Intune administrator + device configuration rights

- **Description: Intune Administrator** (or a least-privilege role that can create device

configuration profiles and assignments).

- **Why required:** You create the Custom OMA-URI profile and assign it to pilot/production groups.
- **How to verify:** Sign in to **Intune admin center** (`intune.microsoft.com`; GCC High:

`intune.microsoft.us`) → **Devices** → **Configuration**.

- **How to implement if missing:** Have an admin grant the role.

3. A printer/driver compatibility inventory

• **Description:** A per-device inventory of **installed print drivers** (V3/V4, third-party vs the Microsoft IPP Class Driver) and **printers**, plus each printer's **Mopria/IPP** and **scan (eSCL)** readiness.

• **Why required:** WPP **removes** incompatible printers and their drivers. You must know what would be removed **before** you enable it.

• **How to verify:** Run `WPPRINT-Get-PrintDriverInventory.ps1` and `WPPRINT-Test-MopriaReadiness.ps1` across a representative sample (and `WPPRINT-Test-ScanCapability.ps1` where multifunction scanning matters).

• **How to implement if missing:** Run the assessment scripts first; that inventory drives the whole plan.

4. Mopria-certified / IPP-capable printers (with IPP enabled)

• **Description:** Target printers must be **Mopria-certified** and reachable over **IPP/IPPS**, so the Microsoft IPP Class Driver can print to them driverlessly.

• **Why required:** Under WPP the PC prints **only** through Windows Ready Print to compatible printers.

• **How to verify:** Check the vendor's Mopria certification and that **IPP Printing / IPPS Printing** are **enabled** in the printer's own settings; `WPPRINT-Test-MopriaReadiness.ps1` probes reachability.

• **How to implement if missing:** Enable IPP/IPPS on the device, replace non-Mopria hardware, or route it through **Universal Print**; devices that can't comply are **excluded** from WPP scope.

5. A target printing path (Ready Print direct and/or Universal Print)

• **Description:** A decision, per printer population, between **direct IPP/Mopria** and **Universal Print** (Microsoft cloud printing) as the modern replacement for legacy print-server/Point-and-Print delivery.

• **Why required:** WPP blocks third-party Point-and-Print; you need the modern path in place first.

• **How to verify:** Universal Print licensing/connector status (if used); direct IPP reachability from the client subnets.

• **How to implement if missing:** Stand up **Universal Print** (or confirm direct IPP) for the printers that need it before removing the legacy path. **Sovereign clouds: confirm Universal Print availability.**

6. Pilot group + change approval

• **Description:** A small, representative **pilot ring** (mix of printer models, at least one multifunction/scanner) and an approved change record.

• **Why required:** WPP changes printing for **every user** on a device; it is staged, not flipped.

• **How to verify:** Pilot Entra group created; change ticket approved naming the assessment findings.

• **How to implement if missing:** Build the pilot group and raise the change with the assessment output.

7. Windows PowerShell for assessment and validation

• **Description:** The included scripts (Windows PowerShell 5.1+) inventory drivers, test Mopria/IPP and scan readiness, read WPP state, and validate the deployment.

• **Why required:** They turn compatibility and post-deployment validation into evidence.

• **How to verify:** `$PSVersionTable.PSVersion`; the scripts run elevated on the target devices.

• **How to implement if missing:** Run them via an Intune **platform script** / Proactive Remediation, or locally elevated on sample devices.

> ***Category-wide prerequisites*** *common to every Intune solution are listed in* `global-requirements-intune-solutions.md`*. Vendor-side printer configuration (enabling IPP/IPPS, confirming Mopria certification) and Universal Print connector setup are referenced, not rebuilt.*

Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** solution in the **Intune Solutions** category. They are published once at the category level so individual listings stay small. Confirm these before deploying *any* Intune solution. Each entry: **Description · Why · Verify · Implement-if-missing.**

G1. Microsoft Entra ID tenant

Description: An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

G2. Microsoft Intune subscription

Description: An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, and security policies. **Verify:** `https://intune.microsoft.com` loads; **Tenant administration** → **Tenant status** shows active. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

G3. Administrative roles

Description: Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies and enrollment settings requires elevated rights. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator**. **Implement:** Assign **Intune Administrator**; reserve Global Administrator for tenant-wide changes.

G4. Device enrollment in Intune

Description: Targeted Windows devices are enrolled and manageable. **Why:** Policies only reach enrolled devices. **Verify:** Intune → **Devices** → **All devices**; the device appears and is **compliant**. **Implement:** Enable **Automatic Enrollment** and join devices to Entra ID.

G5. Pilot security group

Description: A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every solution is piloted before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists. **Implement:** Entra → **Groups** → **New group** → **Security** → add members.

G6. Supported Windows baseline

Description: Windows 10 21H2+ or Windows 11 with current updates. **Why:** Modern Intune features assume a patched, supported OS. **Verify:** `winver`, or Intune → device → **Hardware**. **Implement:** Deploy **Update rings** or upgrade to Windows 11.

G7. Network access to Microsoft cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

The standard deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.

Implementation — Deployment Strategy

The step-by-step plan: assess compatibility, design the target print path, build the Intune profile, pilot and validate, migrate off legacy print servers, roll out in rings, and prevent regression. Portal paths use the commercial hosts; for **GCC High / DoD** use the `*.microsoft.us` equivalents.

Phase 1 — Compatibility assessment (do this first, always)

WPP **removes** incompatible printers and drivers, so the assessment is not optional — it is the safety gate.

Step 1.1 — Inventory drivers and printers

1. Run `WPPRINT-Get-PrintDriverInventory.ps1` across a representative device sample (via an Intune

platform script or Proactive Remediation). It lists installed **print drivers** (V3/V4, third-party vs the **Microsoft IPP Class Driver**) and **printers**, flagging every printer that relies on a **third-party driver** — i.e., what WPP would remove.

Step 1.2 — Test driverless (Mopria/IPP) readiness

1. Run `WPPRINT-Test-MopriaReadiness.ps1`. For each printer it checks reachability over **IPP/IPPS**

and whether the **Microsoft IPP Class Driver** is present, classifying printers as **Ready-Print eligible** or **blocker**.

Step 1.3 — Test scan (eSCL) readiness

1. Where multifunction **scanning** matters, run `WPPRINT-Test-ScanCapability.ps1` to flag scanners

that may lose function under WPP unless they support **Mopria/eSCL**.

Step 1.4 — Read current WPP state and OS eligibility

1. Run `WPPRINT-Get-WPPState.ps1` to record each device's **OS build** (24H2+?) and current WPP state (enabled / disabled / not configured).

Output of Phase 1: a compatibility matrix — for every printer population: *Ready-Print eligible*, *needs Universal Print*, or *blocker (non-Mopria / legacy software / eSCL-less scanner)*.

Phase 2 — Design the target print path

1. **Ready-Print-eligible printers** → keep, printed via the **Microsoft IPP Class Driver** (enable IPP/IPPS on the device where needed).

1. **Printers better served from the cloud** → **Universal Print** (confirm licensing/connector; confirm sovereign-cloud availability).

1. **Blockers** → decide per population: enable IPP/Mopria, replace hardware, keep the scanner's eSCL path, or **exclude those devices from WPP scope** and track them as exceptions.

1. **Legacy "software" printers** (vendor PDF/"print to" utilities) → identify replacements (e.g., the built-in *Microsoft Print to PDF*).

Output of Phase 2: the target design and the **WPP scope** (which device rings get WPP; which are exceptions until remediated).

Phase 3 — Build the Intune configuration profile (Custom OMA-URI)

Because `ConfigureWindowsProtectedPrint` is **not yet in the Settings Catalog**, use a **Custom** profile.

Step 3.1 — Create the profile

1. **Intune admin center** → **Devices** → **Configuration** → **Create** → **New Policy**.
2. **Platform:** Windows 10 and later. **Profile type: Templates** → **Custom**. Click **Create**.
3. **Name:** `WPP - Windows Protected Print (Pilot)`; add a description referencing this deployment.

Step 3.2 — Add the OMA-URI setting

1. **Add** a row:

- **Name:** `Configure Windows Protected Print`
- **OMA-URI:** `./Device/Vendor/MSFT/Policy/Config/Printers/ConfigureWindowsProtectedPrint`
- **Data type: String**
- **Value:** `<enabled/>` *(to turn WPP off later, use `<disabled/>`)*

1. Save the row.

Step 3.3 — Scope to eligible devices only

1. **Assignments:** assign to the **pilot ring** group.
2. Add an **Intune filter** so it applies **only to Windows 11 24H2+** devices (filter on `osVersion`), so ineligible devices are never targeted.
1. Create. The setting writes `WindowsProtectedPrintGroupPolicyState` under `HKLM\Software\Policies\Microsoft\Windows NT\Printers\WPP`.

Phase 4 — Pilot and validate

1. Deploy to the **pilot ring**. After the devices sync, run `WPPRINT-Validate-Deployment.ps1` on pilot devices — it confirms **WPP is enabled** (registry), that **no third-party drivers remain loadable**, and that a **test printer prints via the IPP class driver**.
1. **Manually validate** on at least one multifunction device: print **and scan**. Confirm the removed printers reinstall cleanly via Windows Ready Print.
1. Capture **what got removed** on pilot devices (from the Phase 1 inventory vs the post-deploy state) and confirm it matches the plan — no surprises.
1. Hold the pilot long enough to catch periodic workflows (month-end print jobs, label printers, etc.).

Phase 5 — Migrate off legacy print servers

1. For printers still delivered by **print server + Point and Print + vendor driver**, cut them over to the **modern path** chosen in Phase 2 (direct IPP/Mopria or **Universal Print**).
1. Remove the **Point-and-Print / vendor-driver GPOs and mapped-printer logic** for migrated populations — WPP will block third-party Point-and-Print anyway, so leaving it in place only causes confusing failures.
1. Retire per-population, validating printing after each cutover. Keep the legacy path for un-migrated exceptions until they're remediated.

Phase 6 — Roll out in rings

1. Expand WPP from pilot to production in **rings** (e.g., ring 1 → ring 2 → broad), each gated on the prior ring's validation.
1. Keep the **OS-version filter** on every assignment.
2. Keep **exception devices** (non-Mopria hardware, eSCL-less scanners not yet replaced) **out of WPP scope** until remediated; track them to closure.
1. Monitor the Intune profile's **deployment status** and help-desk tickets after each ring.

Phase 7 — Validate and prevent regression

1. Re-run `WPPRINT-Get-WPPState.ps1` / `WPPRINT-Validate-Deployment.ps1` fleet-wide to confirm WPP is enabled where intended and printing works.
1. Document the **standard modern print path** so new printers are onboarded as **IPP/Mopria or Universal Print** — never a new third-party driver.
1. Keep the assessment scripts in rotation to catch a non-compliant printer/driver **before** it's deployed, so the fleet doesn't drift back toward third-party drivers.

Testing & Validation

Validation is objective: WPP is enabled where intended, printing works through Windows Ready Print, and nothing was removed that wasn't in the plan.

Test 1 — Baseline inventory captured (pre-deployment)

1. Run `WPPRINT-Get-PrintDriverInventory.ps1` and `WPPRINT-Test-MopriaReadiness.ps1` on the pilot devices before assignment.
1. **Pass criteria:** you have a per-device list of printers/drivers and a Ready-Print eligibility classification — the "before" picture WPP will change.

Test 2 — OS eligibility confirmed

1. Run `WPPRINT-Get-WPPState.ps1`.
2. **Pass criteria:** every targeted device reports **Windows 11 24H2+**; ineligible devices are excluded by the assignment filter.

Test 3 — Profile applied, WPP enabled

1. After the pilot syncs, run `WPPRINT-Get-WPPState.ps1` (or `WPPRINT-Validate-Deployment.ps1`).

2. **Pass criteria:** `WindowsProtectedPrintGroupPolicyState` is set under `HKLM\Software\Policies\Microsoft\Windows NT\Printers\WPP`; state = **enabled**; Intune shows the profile **Succeeded**.

Test 4 — Printing works via the IPP class driver

1. On a pilot device, print a test page to a Mopria/IPP printer.

2. **Pass criteria:** the job prints, and the printer is using the **Microsoft IPP Class Driver** (not a vendor V3/V4 driver). `WPPRINT-Validate-Deployment.ps1` confirms.

Test 5 — Removals match the plan

1. Compare the Phase-1 inventory to the post-deployment state.

2. **Pass criteria:** the printers/drivers removed are exactly those the assessment flagged as third-party — **no unplanned removals**. Any Mopria printer reinstalls via Windows Ready Print.

Test 6 — Third-party Point and Print is blocked

1. Attempt to install a printer that would pull a **third-party driver** via Point and Print.

2. **Pass criteria:** the third-party driver install is **blocked** — confirming the WPP hardening.

Test 7 — Scanning validated (multifunction devices)

1. On an MFD in scope, perform a scan.

2. **Pass criteria:** scanning works via **Mopria/eSCL**, or the device is correctly identified as a scan **exception** and kept out of WPP scope until remediated.

Test 8 — No collateral impact

1. Confirm eligible non-pilot devices are unaffected, and exception devices retained their required (non-Mopria) printers because they were excluded from scope.

1. **Pass criteria:** only the intended ring changed; exceptions behave as before.

Evidence to keep

- The pre/post inventory diff (what WPP removed vs. the plan).
- The WPP-enabled state and a successful IPP-class-driver print.
- The scan validation and the exception list.

Troubleshooting Guide

Symptom → most likely cause → verify → fix, for a Windows Ready Print / WPP deployment through Intune.

A. The Custom OMA-URI profile shows "Error" or "Not applicable"

Cause	Verify	Fix
Device isn't **Windows 11 24H2+**	`winver` / `WPPRINT-Get-WPPState.ps1`	Update to 24H2, or scope the profile with an OS-version filter so ineligible devices aren't targeted
Unsupported **edition**	`Get-ComputerInfo OsName`	WPP supports Pro/Enterprise/Education/IoT Enterprise — Home is not supported
OMA-URI/ data type wrong	Re-open the row	Must be exactly `./Device/Vendor/MSFT/Policy/Config/Printers/ConfigureWindowsProtectedPrint`, **String**, value `<enabled/>`

*"Not applicable" is almost always the **24H2 gate** — add the filter and it resolves.*

B. A printer disappeared after WPP was enabled

- **Expected if** it relied on a **third-party driver** — WPP removes those printers and their drivers.
- **Verify:** it's on the Phase-1 third-party list; `WPPRINT-Get-PrintDriverInventory.ps1` before/after.
- **Fix:** reinstall it as a **Mopria/IPP** printer via Windows Ready Print, or add it to **Universal

Print**. If it truly can't go driverless, that** device **belongs in the** exception** set (out of WPP scope) until the hardware is replaced.

C. Can't print at all on a WPP device

- **Likely cause:** the printer isn't reachable over **IPP/IPPS**, or **Mopria/IPP isn't enabled** on the

printer, so the IPP class driver has nothing to bind to.

- **Verify:** `WPPRINT-Test-MopriaReadiness.ps1`; confirm **IPP Printing / IPPS Printing** are enabled in

the printer's own settings; confirm network reachability to the printer.

- **Fix:** enable IPP/IPPS on the device, correct the network path, or route via Universal Print.

D. Scanning stopped working on a multifunction device

- **Likely cause:** the scanner relied on a vendor driver / WSD; WPP's driverless model needs

Mopria/eSCL scan support.

- **Verify:** `WPPRINT-Test-ScanCapability.ps1`.
- **Fix:** use the device's **eSCL** scan path if supported; if not, keep that device **out of WPP scope**

until the scanner is replaced with an eSCL-capable model. Don't enable WPP fleet-wide over a scanner gap — scope around it.

E. A legacy "software" printer (vendor PDF/"print to" tool) is gone

- **Expected:** WPP can remove legacy software printers.
- **Fix:** use the built-in **Microsoft Print to PDF**, or a Ready-Print-compatible replacement. Update

any workflow/documentation that named the old software printer.

F. Point-and-Print printer install now fails

- **Expected under WPP:** third-party Point and Print is **blocked** by design.
- **Fix:** deliver the printer through the **modern path** (direct IPP/Mopria or Universal Print) instead

of print-server Point and Print. Remove the legacy mapped-printer/GPO logic for migrated populations.

G. WPP didn't change anything / you want to revert

- **Verify:** `WPPRINT-Get-WPPState.ps1` — is the registry value actually set?
- **Revert:** set the OMA-URI value to `<disabled/>` (or remove the assignment). Note that reverting does

not automatically restore removed third-party drivers — they must be reinstalled by the modern path or, where unavoidable, redeployed deliberately.

H. Sovereign cloud (GCC High / DoD)

- **Symptom:** portal paths or Universal Print options don't match.
- **Fix:** use `intune.microsoft.us`; the CSP and method are identical. **Universal Print availability

differs by sovereign cloud** — confirm for yours before designing a cloud-print path.

Escalation

Open a Microsoft support case only after: the profile applies but WPP state won't set on a confirmed 24H2 device; a Mopria-certified printer with IPP enabled still won't print via the class driver; or a suspected platform issue. Attach the WPP-state output, the driver inventory, and the profile configuration.

Best Practices

Assess before you enforce

- **Never enable WPP fleet-wide before the compatibility assessment.** WPP removes incompatible printers

and their drivers; the assessment (drivers, Mopria/IPP, eSCL scan) is what turns that from a production surprise into a planned change.

- Treat the **pre/post inventory diff** as the acceptance test — the only removals should be the ones the

assessment predicted.

Scope deliberately

- WPP is **device-scoped** — it affects **every user** on the PC. Plan shared/kiosk/lab devices with that

in mind.

- Always attach an **OS-version filter (24H2+)** to every assignment so ineligible devices are never

targeted (and don't show spurious "Not applicable" errors).

- Keep **non-compliant devices out of scope** (non-Mopria hardware, eSCL-less scanners) and track them as

exceptions to closure — don't hold the whole rollout for a handful of laggards, and don't force WPP over a known gap.

Modern path first, then harden

- Stand up the **modern print path** (direct IPP/Mopria or **Universal Print**) **before** removing the

legacy print-server/Point-and-Print path. Enabling WPP with no replacement path in place is how you create "can't print at all."

- Enable **IPP/IPPS on the printers** and confirm **Mopria certification** as part of readiness — the

client side is only half of driverless printing.

Roll out in rings

- Pilot → ring 1 → ring 2 → broad, each gated on the prior ring's validation and help-desk signal.

- Hold each ring long enough to catch **periodic** workflows (month-end runs, label/receipt printers,

scan-heavy roles).

Prevent regression

- Document the **standard modern onboarding**: new printers arrive as **IPP/Mopria or Universal Print**,

never as a new third-party driver.

• Keep the **assessment scripts in rotation** (e.g., as a Proactive Remediation) to catch a third-party driver creeping back in before it spreads.

• Align with Microsoft's **V3/V4 driver end-of-servicing** direction — treat any new third-party driver request as an exception that needs justification, not a default.

Evidence & handoff

• Keep the assessment matrix, the pre/post diff, the WPP-enabled proof, and the exception list with the change record.

• Feed confirmed blockers (specific non-Mopria models, eSCL-less scanners) into procurement so replacements are Ready-Print-compatible.

Sovereign clouds

• Use `intune.microsoft.us` and the same CSP; **confirm Universal Print availability in your sovereign cloud** before designing a cloud-print path, and document which path each population uses.

Deliverables Summary

Everything included in this solution.

Documentation (compiled to PDF)

1. **Solution Overview** — Windows Ready Print vs. Windows Protected Print mode, the three-tier architecture, what enabling WPP actually does, and how it's delivered through the Intune Policy CSP.
1. **Requirements** — Windows 11 24H2+, Intune rights, the compatibility inventory, Mopria/IPP printers, a target print path, pilot + change approval, and PowerShell — each with why / verify / implement.
1. **Implementation — Deployment Strategy** — the Phase 1–7 runbook: assess → design the target path → build the Custom OMA-URI profile → pilot and validate → migrate off legacy print servers → roll out in rings → prevent regression.
1. **Testing & Validation** — eight objective tests, from baseline inventory to "removals match the plan" and a successful IPP-class-driver print.
1. **Troubleshooting Guide** — symptom→cause→verify→fix for the Custom-profile errors, removed printers, can't-print, scanning loss, legacy software printers, blocked Point-and-Print, reverting, and sovereign clouds.
1. **Best Practices** — assess before enforce, scope deliberately, modern-path-first, ring rollout, and

regression prevention.

1. **Executive Summary** — the one-page brief for stakeholders.

PowerShell tools (packaged as a ZIP)

1. `WPPRINT-Get-PrintDriverInventory.ps1` — inventories installed print drivers and printers, flagging

third-party (V3/V4) drivers vs. the Microsoft IPP Class Driver — i.e., what WPP would remove.

1. `WPPRINT-Test-MopriaReadiness.ps1` — tests each printer's IPP/IPPS reachability and driverless

(Mopria/IPP class driver) eligibility.

1. `WPPRINT-Test-ScanCapability.ps1` — flags multifunction scanners that may lose function under WPP

unless they support Mopria/eSCL.

1. `WPPRINT-Get-WPPState.ps1` — reports OS build (24H2+ eligibility) and the current WPP state from the

registry.

1. `WPPRINT-Validate-Deployment.ps1` — post-deployment validation: WPP enabled, no third-party drivers

remaining, and a test print via the IPP class driver.

All scripts are Windows PowerShell 5.1+ compatible, run elevated (deployable via Intune platform script / Proactive Remediation), and are parse-verified.

Scope

• **In scope:** deploying and hardening **Windows Ready Print / WPP** through Intune on **Windows 11

24H2+**, with compatibility assessment and legacy print-server migration.

• **Referenced, not rebuilt:** vendor-side printer configuration (enabling IPP/IPPS, Mopria

certification) and Universal Print connector setup.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

WPPRINT-Get-PrintDriverInventory.ps1

How to run: Run elevated on a target device (or deploy via an Intune platform script / Proactive Remediation): .\WPPRINT-Get-PrintDriverInventory.ps1 .\WPPRINT-Get-PrintDriverInventory.ps1 -OutputPath C:\Temp

WPPRINT-Get-WPPState.ps1

How to run: Run on a target device (locally elevated, or via Intune platform script / Proactive Remediation): .\WPPRINT-Get-WPPState.ps1

WPPRINT-Test-MopriaReadiness.ps1

How to run: Run on a target device on the same network as the printers: .\WPPRINT-Test-MopriaReadiness.ps1 .\WPPRINT-Test-MopriaReadiness.ps1 -OutputPath C:\Temp

WPPRINT-Test-ScanCapability.ps1

How to run: Run on a target device on the same network as the MFDs: .\WPPRINT-Test-ScanCapability.ps1 .\WPPRINT-Test-ScanCapability.ps1 -ScannerHosts 10.0.0.50,10.0.0.51

WPPRINT-Validate-Deployment.ps1

How to run: Run elevated on a validated device (or via Intune platform script / Proactive Remediation): .\WPPRINT-Validate-Deployment.ps1 .\WPPRINT-Validate-Deployment.ps1 -TestPrinter "Contoso-MFD-3rdFloor"